

DELIVERY OPERATIONS & PROFITABILITY ANALYTICS

Power BI project report | Snowflake + SQL + DAX + interactive analytics

Project purpose
Analyze simulated delivery operations to understand service performance, warehouse and partner behavior, and profitability - then convert the findings into an interactive Power BI management report.

10,000	2,000	100	50	730
unique orders	customers	delivery partners	warehouses	date records

Headline dashboard metrics

Total Revenue: ₹12.75M | Total Profit: ₹9.40M | Profit Margin: 73.74% | Loss-Making Orders: 3.2% | Late Delivery: 44.36%

Important: all operational and financial figures in this report are generated from a synthetic dataset. They demonstrate analytical design and workflow; they are not claims about a real logistics company.

1. Executive Summary

The project builds an end-to-end analytics workflow for a simulated delivery business. Raw order and dimension tables are loaded into Snowflake, cleaned and transformed with SQL, modeled as a star schema in Power BI, and analyzed with DAX. The final report is designed for management users who need a fast view of delivery performance, profitability, regional differences, warehouse detail, and controlled access.

The analytical design deliberately avoids a pre-supplied delay-reason column. Instead, users are expected to reason from measurable operational stages such as processing days, transit days, partner performance, warehouse-level patterns, distance, and regional comparisons.

Primary management questions

- Why are deliveries late, and where are late deliveries concentrated?
- Which warehouses, regions, and delivery partners show weaker performance?
- Where is the business generating profit, and where are loss-making orders concentrated?
- Does operational performance appear associated with profitability?
- What actions should management investigate or prioritize?

Report deliverables

Power BI dashboard, PDF project report, user guide presentation, and data architecture documentation.

2. Solution Architecture & Data Flow

Synthetic Raw Data CSV source tables 10,020 raw orders + dimensions	Snowflake LOGISTICSDATA / PUBLIC raw landing tables	SQL Cleaning & Transformation standardize, deduplicate, validate, map UNKNOWN	Star Schema FactOrders + four dimensions 1-to-many relationships	Power BI DAX, dashboards, RLS, drill-through, bookmarks
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Technology stack

Layer	Technology / role	Purpose
Source	Synthetic CSV files	Generate repeatable, realistic-looking logistics data with intentional quality issues.
Data platform	Snowflake	Central landing and analytical data store; database LOGISTICSDATA, schema PUBLIC.
Transformation	SQL	Standardize text, deduplicate, validate dates, repair dimensional keys with UNKNOWN mapping.
Semantic model	Power BI star schema	FactOrders connected to customer, partner, warehouse, and date dimensions.
Analytics	DAX	Measures for orders, revenue, profit, margin, loss-making orders, and delivery performance.
Presentation & security	Power BI	Three report pages, bookmarks, slicers, tooltips, drill-through, page navigation, and RLS.

The pipeline preserves a clear separation between raw source data and cleaned analytical tables. Raw data remains available for auditability, while business-facing visuals use cleaned tables and controlled calculations.

3. Data Model & Grain

Fact grain: one row represents one unique customer delivery order. The final cleaned fact table contains 10,000 unique orders.

Star schema

Table	Approx. rows	Key fields / role
FACT_ORDERS	10,000	Order_ID (business key); Customer_ID, Warehouse_ID, Delivery_Partner_ID; dates; financial and operational fields.
DIM_CUSTOMER	2,000	Customer_ID; customer, city/state, segment, status.
DIM_DELIVERY_PARTNER	100	Delivery_Partner_ID; partner, base city, vehicle type, experience, rating.
DIM_WAREHOUSE	50 + UNKNOWN	Warehouse_ID; warehouse, city/state, capacity, region.
DIM_DATE	730	Continuous Date table covering 2024-01-01 to 2025-12-30.

Relationships

DIM_CUSTOMER 1 → * FACT_ORDERS; DIM_DELIVERY_PARTNER 1 → * FACT_ORDERS; DIM_WAREHOUSE 1 → * FACT_ORDERS; DIM_DATE 1 → * FACT_ORDERS using Order_Date.

Key modeling principle

Dimension tables filter the fact table. This enables region, warehouse, customer, partner, and date context to flow consistently into DAX measures and visuals.

4. Data Quality Profiling & SQL Transformation

The synthetic raw data was intentionally built with common real-world issues so the project demonstrates an analyst workflow beyond visualization.

Observed / injected issues

Issue	Handling	Result
Duplicate orders	DISTINCT and final Order_ID deduplication	Final FACT_ORDERS = 10,000 unique orders.
Inconsistent text casing / spaces	TRIM + INITCAP for location/status text	Cities and status categories standardized.
Invalid delivery dates	Actual_Delivery_Date set to NULL when earlier than order/dispatch date	12 affected rows retained; excluded from delivery-time analysis.
Broken foreign keys	Map missing customer/warehouse/partner IDs to UNKNOWN dimension members	Final FK checks return zero invalid references.
Legitimate NULL dates	Preserve missing actual dates for cancelled/failed orders	Cancelled/failed orders remain analyzable without inventing events.
Outliers	Profile numeric values and flag extreme observations such as very large Distance_KM	Outliers remain visible for analytical review rather than being silently deleted.

Business rules

- Cancelled orders are excluded from late-delivery calculations because no completed delivery occurred.
- Late Delivery % = Late Delivered Orders / Valid Delivered Orders.
- Delivered rows require a valid Actual_Delivery_Date for delivery-time metrics.
- Processing Days = Order_Dispatched_Date - Order_Date.
- Transit Days = Actual_Delivery_Date - Order_Dispatched_Date.

A key analytical limitation is that there is no Reason_for_Delay column. Root-cause analysis therefore relies on relationships and patterns among measurable operational variables rather than a pre-labeled cause.

5. DAX & Analytical Calculations

DAX was used for report-level calculations so the same definitions respond dynamically to filters, bookmarks, and drill-through context.

Core measures

Measure	Definition
Total Orders	COUNTROWS(FACT_ORDERS)
Total Revenue	SUM(FACT_ORDERS[Order_Value])
Total Profit	Revenue - Freight - Packaging - Warehouse Handling - Delivery Partner Cost
Profit Margin %	DIVIDE([Total Profit], [Total Revenue])
Loss-Making Orders	Orders where Order_Value - all listed order-level costs < 0
Loss-Making Orders %	DIVIDE([Loss-Making Orders], [Total Orders])
Late Delivered Orders	Delivered orders where Actual_Delivery_Date > Promised_Delivery_Date
Valid Delivered Orders	Delivered orders with non-blank Actual_Delivery_Date
Late Delivery %	DIVIDE([Late Delivered Orders], [Valid Delivered Orders])
Average Processing Days	AVERAGE(FACT_ORDERS[Processing Days])
Average Transit Days	AVERAGE(FACT_ORDERS[Transit Days])

Calculated columns

Processing Days and Transit Days are stored at order grain because each order needs its own duration. Measures then aggregate those row-level values under the active filter context.

Analytical principle

A calculated column answers “what is the value for this order?” A measure answers “what is the aggregate for the current filter context?” This distinction keeps the model efficient and the dashboard responsive.

6. Dashboard Page 1 - Executive Overview

Purpose: give management a one-screen summary of order volume, revenue, profit, margin, loss-making orders, late delivery, regional distribution, warehouse delay concentration, and order-status mix.

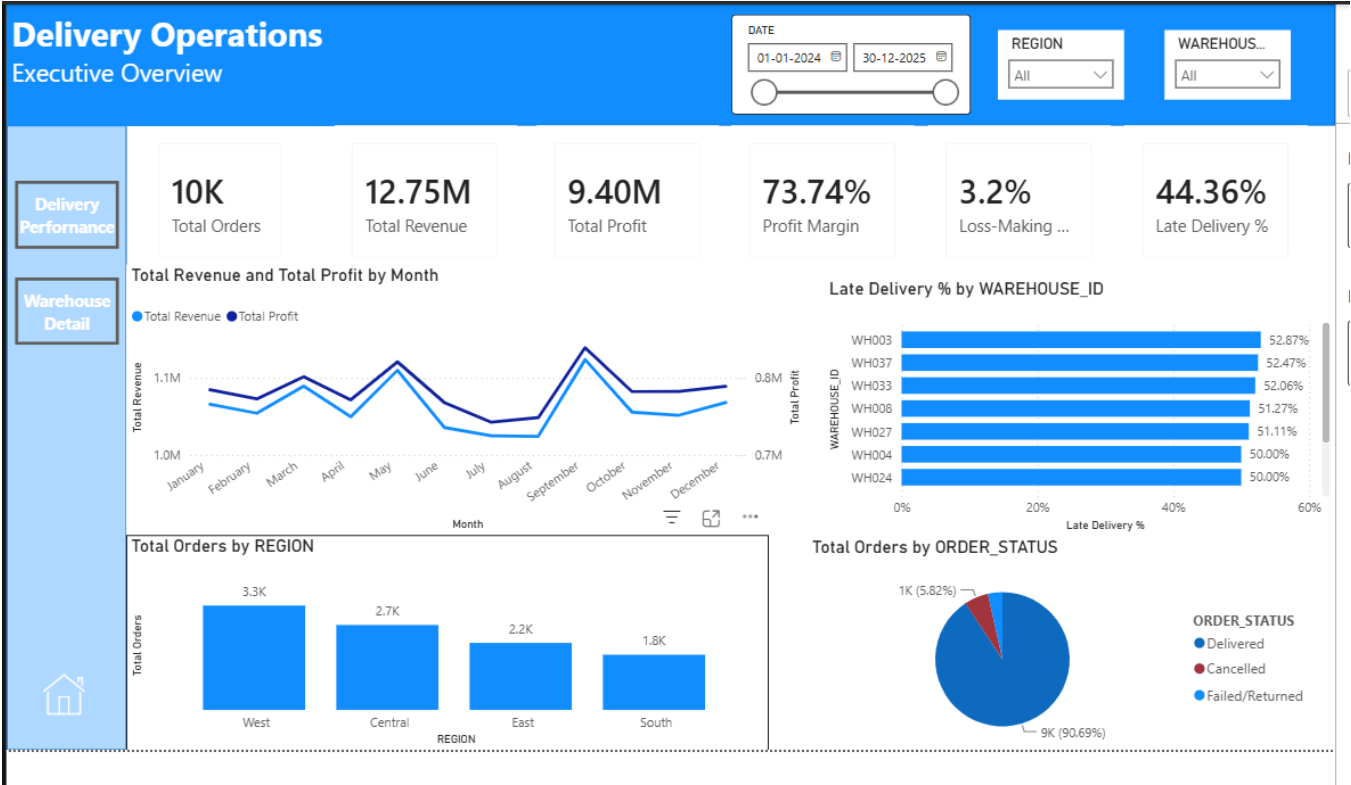


Figure 1. Executive Overview screenshot from the completed Power BI report.

Interaction	How it works
Date slicer	Filters the report to the selected date interval.
Region slicer	Filters all applicable visuals to the selected warehouse region.
Warehouse slicer	Focuses the report on a chosen warehouse.
Page navigation	Left-side buttons navigate to Executive Overview, Delivery Performance, and Warehouse Details.
Visual interaction	Selecting a category in a chart cross-filters/highlights related visuals where applicable.

7. Dashboard Page 2 - Delivery Performance & Profitability

Page 2 uses bookmarks to provide two analytical views without creating a separate page: Operational Analysis and Profitability Analysis.



Figure 2. Profitability Analysis bookmark state.

Operational Analysis

The operational view contains KPI cards for processing time, transit time, late delivered orders, and late delivery %, plus warehouse-level analysis, a processing-vs-late scatter, partner transit analysis, and late delivery % by region.

Profitability Analysis

The profitability view contains revenue, total profit, profit margin, and loss-making order KPIs, with warehouse, region, customer-segment, and monthly profit visuals.

Bookmark behavior

- Operational Analysis button loads the operational visual state.
- Profitability Analysis button loads the profitability visual state.
- The two states share the same page and report filters, so users can switch analytical perspective quickly.

8. Dashboard Page 3 - Warehouse Details & Drill-through

Purpose: move from portfolio-level analysis into a specific warehouse. The page is configured with Warehouse_ID as a drill-through field, so all visuals respond to the selected warehouse.

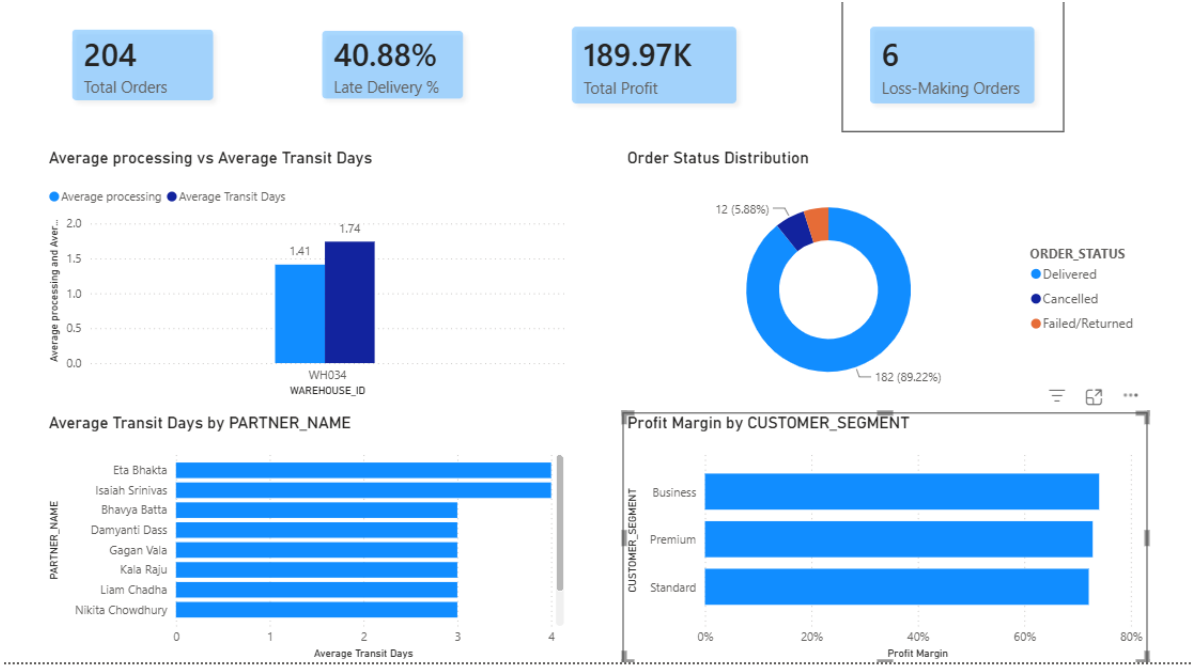


Figure 3. Warehouse Details screenshot for a selected warehouse.

Element	Warehouse Detail behavior
KPI cards	Total Orders, Late Delivery %, Total Profit, and Loss-Making Orders are recalculated for the selected warehouse.
Processing vs Transit	Compares the warehouse average processing and transit durations.
Order Status Distribution	Shows delivered, cancelled, and failed/returned outcomes for the selected warehouse.
Delivery Partner chart	Compares average transit days for partners associated with the warehouse.
Profit Margin by Segment	Shows margin differences across customer segments within the warehouse context.
Back button	Returns the user to the page from which drill-through was initiated.

Drill-through workflow

Right-click a warehouse in an eligible visual → Drill through → Warehouse Details. Power BI transfers the Warehouse_ID filter to Page 3. “Keep all filters” is enabled, preserving relevant report context.

9. Interactivity, Tooltips & Row-Level Security

Custom tooltip

A dedicated report-page tooltip provides four warehouse metrics in a compact table: Total Orders, Average Processing Days, Average Transit Days, and Late Delivery %. Hovering over a warehouse in the relevant visual reveals those metrics for the hovered warehouse.

Row-Level Security (RLS)

Region-specific roles were created so users can be restricted to a regional slice of the warehouse/fact model. The implementation relies on the warehouse dimension filtering FACT_ORDERS through the existing one-to-many relationship.

Example role: south limits DIM_WAREHOUSE[Region] to South. Other roles were also tested for Central, East, and West. The roles were validated in Power BI using “View as roles”.

Security design principle

RLS is applied at the dimension layer instead of directly duplicating filters on the fact table. This keeps the model aligned with star-schema filtering behavior and makes the security rule easier to maintain.

Other navigation features

- Home / page-navigation buttons provide a consistent route between report pages.
- Bookmark controls switch analytical views on Page 2.
- The Warehouse Details page contains a Back control generated for drill-through navigation.

10. Key Findings & Management Interpretation

The values below are observations from the completed synthetic report. They should be described as patterns or associations, not causal proof.

Executive-level findings

- The report contains 10,000 unique orders with approximately ₹12.75M of order value/revenue and ₹9.40M of modeled profit, producing a 73.74% margin under the project cost definition.
- Late Delivery % is approximately 44.36%, indicating that delivery timeliness is a major area for operational investigation in the simulated dataset.
- Loss-making orders are approximately 3.2% of total orders, so the profitability risk is concentrated in a minority of orders.

Regional profitability

- West is the highest-profit region at about ₹3.2M, followed by Central at about ₹2.6M, East at about ₹2.0M, and South at about ₹1.7M.

Customer-segment risk

- Loss-making order rate is highest for Standard (~4.4%), followed by Premium (~4.2%) and Business (~1.9%). This is a risk concentration signal, not evidence that segment membership causes losses.

Monthly trend

- Monthly profit varies through the year in the completed report, with visible peaks and troughs rather than a flat pattern. The chart describes when performance changed; additional analysis is needed to explain why.

Warehouse drill-through

- Warehouse-level views expose differences in processing, transit, partner performance, order outcomes, and customer-segment margin after filtering to a single warehouse.

11. Recommendations & Next Analytical Steps

Operational recommendations

- Investigate warehouses with persistently high late-delivery percentages together with processing and transit metrics.
- Review partner-level transit performance inside underperforming warehouses instead of assuming the partner alone is the cause.
- Compare regional performance while accounting for order mix and volume so management does not confuse scale with efficiency.

Profitability recommendations

- Investigate the small population of loss-making orders individually; this is likely to yield more actionable findings than broad averages.
- Examine Standard-segment loss-making orders for differences in order value, distance, cost components, and warehouse/partner mix.
- For high-profit regions and warehouses, examine whether profitability is driven by volume, order value, lower cost, or a combination.

Limitations

- The dataset is synthetic and not representative of a real company or external market conditions.
- No explicit delay-cause field exists, so the dashboard cannot directly prove causal root causes.
- Order-level cost modeling does not allocate fixed overhead such as warehouse rent or office electricity.
- Monthly date views must retain year context; grouping only by month name can combine different years.

Recommended future enhancement

Add a governed reason-code dataset or operational event log (for example, dispatch queue, route exceptions, customer-not-home, weather, or capacity events) so future root-cause analysis can move from association toward stronger causal investigation.

12. Validation Checklist & Interview Summary

Final QA checklist

- Fact table contains 10,000 unique Order_ID values.
- Broken foreign keys are mapped to UNKNOWN dimension members; final invalid FK count is zero.
- Invalid actual-delivery dates are nullified without deleting the corresponding order rows.
- Date, region, and warehouse slicers work on the Executive Overview.
- Page navigation buttons work across all report pages.
- Operational and Profitability bookmarks switch correctly on Page 2.
- Warehouse drill-through lands on Page 3 with selected Warehouse_ID filter.
- RLS roles were tested with View as roles.
- Custom warehouse tooltip displays filtered metrics.

60-second interview explanation

"I built a synthetic delivery-operations analytics solution end to end. I generated raw order and dimension data with realistic quality issues, loaded it into Snowflake, cleaned and validated it with SQL, and modeled the final data as a star schema in Power BI. I then built DAX measures for revenue, profit, margin, loss-making orders, processing time, transit time, and late delivery. The report has an executive page, an operational/profitability analysis page with bookmarks, and a warehouse drill-through page. I also implemented custom tooltips, page navigation, and region-based RLS, and validated the interactions using Power BI's testing features."

End of report.